



Hardest known substance

Researchers calculated the strength of Nuclear Pasta found deep inside the crust of Neutron stars using simulations. <https://digit.in/oct18-74-1>



Super-Earth found

The NASA's new planet hunter found an evaporating Super-Earth orbiting Pi Mensae 60 light-years from Earth. <https://digit.in/oct18-74-2>

The Crew module Atmospheric Reentry Experiment

ogy of the entire country. Every one of the Indians can be proud that India is on par with the other developed countries."

With the Gaganyaan mission, ISRO hopes to make India the fourth country in the world to realise manned spaceflight. The other countries with a manned space program so far are Russia, the United States of America and China. Technically, India has already been part of a manned spaceflight mission. On Monday, 2 April, 1984, Rakesh Sharma, a pilot of the Indian Air Force, flew on board the Soyuz T-11 launch vehicle to the Soyuz T-11 space station.

There have been three flights that have been significant steps towards the preparation of Gaganyaan, even before the mission was officially greenlit and named. The first was the Space Capsule Recovery Experiment, which demonstrated ISRO's capability of sending an orbital platform to space, controlling it, and recovering it. The module was successfully recovered after spending 10 days in space, in January 2007. The learnings from this experiment was applied at another mission, with an upgraded crew module. The Crew module Atmospheric Reentry Experiment (CARE) mission blasted off in December 2014. The CARE module separated from the rocket 330 seconds after lift off, and began a guided descent into the atmosphere, successfully returning to the Andaman Sea in the Bay of Bengal. In July 2018, ISRO qualified its Crew Escape System. In case something goes wrong during the launch of the manned mission, and the spaceflight has to be aborted, the crew escape system ensures that the crew module separates from the rocket, deploys the parachutes, and returns the personnel safely back to the Earth. The test was successfully conducted in under 259 seconds.

THE SPACEFLIGHT

The launch vehicle for the first manned spaceflight is expected to be the Geosynchronous Satellite

GAGANYAAN

India's first human mission to space has been greenlit, and the preparations are on in full swing to unfurl the Indian flag in space before the country turns 75.

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he Indian space program was a big part of Prime Minister Narendra Modi's address on 15

August 2018. The IRNSS navigation constellation, the Mangalyaan mission and the reliable satellite launches were all celebrated over the course of his speech. Then, the Prime Minister went on to make a blockbuster announcement. The address was given in Hindi, but the English rendering is, "we have resolved that by 2022, when India celebrates 75 years of Independence or maybe even before that, certainly some of our young boys and girls will unfurl the Tricolour in space. With Mangalyaan our scientists have proved their capabilities. I feel proud to

announce that very soon as a part of our Manned-Space mission, we shall be sending an Indian into space." On 28 August, in a press conference about issues related to the Department of Space, ISRO chairman Dr K Sivan provided more details about the ambitious mission, including the name of the mission, "Gaganyaan". Dr Sivan made it a point to note that as far as ISRO was concerned, that this was a "human mission", and the crew could consist of either men or women. Just a few days later, at the 6th Bengaluru Space Expo, the star attraction of the show was a crew module and an indigenously developed "flightsuit" for the astronauts.

THE PREPARATION

Considering the deadline for the mission, Sivan said it was a "very very tight schedule." He went on to explain, "This program is really going to enhance the science and technol-